EX 8.2.1

	1	2	3	SUPPLY
1	15 7	9	13	7
2	11	M 3	17 2	5
3	9	11	9 3	3
DEMAND	7	3	5	

EX 8.2.2

	1	2	3	4	5	SUPPLY
1	2 4	4	6	5	7	4
2	7	6 4	3 2	M	4	6
3	9	7	5	2 5	5 1	6
4	0	0	0	0	0 4	4
DEMAND	4	4	2	5	5	

Ex 8.2.6

	1	2	3	4	5	SUPPLY	u_i
1	8 / 1	6 / 3	3 / 20	7 / 1	5 / 2	20	2
2	5 / 25	M / M-1	8 / 7	4 / 5	7 / 6	30	0
3	6 / -1	3 / 25	9 / 6	6 / 5	8 / 5	30	2
4	0 E / -4	0 L / 0	0 / 0	0 / -3	0 / 20	20	-1
DEMAND	25	25	20	10	20		
v_j	5	1	1	4	1		

$z = 125 + 75 + 60 + 20 + 30 = 310$

Filling the blanks: $u_2 = 0$ $v_1 = 5$ $v_4 = 4$ $u_3 = 2$ $v_2 = 1$ $u_4 = -1$ $v_3 = 1$ $v_5 = 1$ $u_1 = 2$

E : Entering
L : Leaving

	1	2	3	4	5	Supply	u_i
1	8 / 5	6 / 7	3 / 20	7 / 5	5 / 2	20	3 (3)
2	5 / 25	M / M-1	8 / 3	4 / 5	7 / 2	30	5 (3)
3	6 E / -1	3 / 25	9 / 2	6 L / 5	8 / 1	30	7 (5)
4	0 / 0	0 / 4	0 / 0	0 / 1	0 / 20	20	0 (1)
DEM.	25	25	20	10	20		
v_j	0 (2)	-4 (8)	0 (2)	-1 (4)	0 (2)		

$z = 125 + 75 + 60 + 20 + 30 = 310$

GREEN NUMBERS: ORDER FOR UPDATE u_i and v_j

	1	2	3	4	5	SUPPLY	u_i
1	8 / 5	6 / 6	3 / 20	2 / 5	5 / 2	20	3^{3}
2	5 / 20	M / M-1	8 / 3	4 / 10	7 / 2	30	5^{3}
3	6 / 5	3 / 25	8 / 3	6 / 1	8 / 2	30	6^{3}
4	0 / 0	0 / 3	0 / 0	0 / 1	0 / 20	20	0^{1}
DEMAND	25	25	20	10	20	$Z = 305$	
v_j	0^{2}	-3^{4}	0^{2}	-1^{5}	0^{2}		

SOL IS OPTIMAL! (NO NEGATIVE $c_{ij} - u_i - v_j$)

8.2.8 (a, b)

a.

		DESTINATION				
		1	2	3	4	SUPPLY
SOURCE	1	700	800	500	200	10
	2	200	900	100	400	20
	3	400	500	300	100	20
	4	200	100	400	300	10
DEMAND		20	10	10	20	

b.

	1	2	3	4	SUPPLY	u_i
1	700 10	800	500	200	10	
2	200 10	900 10	100 0	400	20	
3	400	500	300 10	100 10	20	
4	200	100	400	300 10	10	
Demand	20	10	10	20		
v_j						

in case you want to do .c : the OPTIMAL SOLUTION is

$x_{14} = 10$ $\quad x_{21} = 20$ $\quad x_{33} = 10$ $\quad x_{34} = 10$ $\quad x_{42} = 10$

$Z = 11.000$

Exercise 8.2-8(c)

Initial BFS

	1	2	3	4	Supply	u_i
1	700 / 10	800 / $\cdots$	500 / $\cdots$	200 / c_{12}	10	a_1
2	200 / 10	900 / 10	100 / ..	400 / $\cdots$	20	a_1
3	400 / $\cdots$	500 / 0	300 / 10	100 / 10	20	a_1
4	200 / $\cdots$	100 / $\cdots$	400 / $\cdots$	300 / 10	10	a_1
Demand	20	10	10	20		
v_j	b_1	b_2	$\cdots$	b_n		

Iteration 1

	1	2	3	4	Supply	u_i
1	700 L / 10	800 / -600	500 / -700	200 E / -800	10	700
2	200 / 10	900 / 10	100 / -600	400 / -100	20	200
3	400 / 600	500 / 0	300 / 10	100 / 10	20	-200
4	200 / 200	100 / -600	400 / -100	300 / 10	10	0
Demand	20	10	10	20		
v_j	0	700	500	300		

Z=25000

Iteration 2

	1	2	3	4	Supply	u_i
1	700 / 800	800 / 200	500 / 100	200 / 10	10	100
2	200 / 20	900 L / 0	100 E / -600	400 / -100	20	400
3	400 / 600	500 / 10	300 / 10	100 / 0	20	0
4	200 / 200	100 / -600	400 / -100	300 / 10	10	200
Demand	20	10	10	20	Z=17000	
v_j	-200	500	300	100		

Iteration 3

	1	2	3	4	Supply	u_i
1	700 / 200	800 / 200	500 / 100	200 / 10	10	100
2	200 / 20	900 / 600	100 / 0	400 / 500	20	-200
3	400 / 0	500 L / 10	300 / 10	100 / 0	20	0
4	200 / -400	100 E / -600	400 / -100	300 / 10	10	200
Demand	20	10	10	20	Z=17000	
v_j	400	500	300	100		

Iteration 4

	1	2	3	4	Supply	u_i
1	$\boxed{700}$ 200	$\boxed{800}$ 800	$\boxed{500}$ 100	$\boxed{200}$ 10	10	200
2	$\boxed{200}$ 20	$\boxed{900}$ 1200	$\boxed{100}$ 0	$\boxed{400}$ 500	20	−100
3	$\boxed{400}$ 0	$\boxed{500}$ 600	$\boxed{300}$ 10	$\boxed{100}$ 10	20	100
4	$\boxed{200}$ E −400	$\boxed{100}$ 10	$\boxed{400}$ −100	$\boxed{300}$ L 0	10	300
Demand	20	10	10	20		Z=11000
v_j	300	−200	200	0		

Iteration 5

	1	2	3	4	Supply	u_i
1	$\boxed{700}$ 200	$\boxed{800}$ 400	$\boxed{500}$ 100	$\boxed{200}$ 10	10	300
2	$\boxed{200}$ 20	$\boxed{900}$ 800	$\boxed{100}$ 0	$\boxed{400}$ 500	20	0
3	$\boxed{400}$ 0	$\boxed{500}$ 200	$\boxed{300}$ 10	$\boxed{100}$ 10	20	200
4	$\boxed{200}$ E 0	$\boxed{100}$ 10	$\boxed{400}$ 300	$\boxed{300}$ L 400	10	0
Demand	20	10	10	20		Z=11000
v_j	200	100	100	−100		

Optimal Solution: $x_{14} = 10, x_{21} = 20, x_{33} = 10, x_{34} = 10, x_{42} = 10$, cost: $\$11,000$